

plancher@dartmouth.edu
Belmont, MA

BRIAN K. PLANCHER

brianplancher.com a2r-lab.org

 [plancherb1](https://orcid.org/plancherb1)  [a2r-lab](https://www.researchgate.net/profile/a2r-lab)
 [user=Hys8HdsAAAAJ](https://github.com/Hys8HdsAAAAJ)
 [0000-0002-0078-3653](https://doi.org/10.0000-0002-0078-3653)

ACADEMIC POSITIONS

DARTMOUTH COLLEGE: Hanover, NH

- Assistant Professor of Computer Science
- Visiting Scholar

SEPT 2025 – PRESENT
APRIL 2025 – AUG 2025

BARNARD COLLEGE, COLUMBIA UNIVERSITY: New York, NY

- Research Scientist
- Assistant Professor of Computer Science

JULY 2025 – DEC 2025
JULY 2022 – JUNE 2025

FU FOUNDATION SCHOOL OF ENGINEERING AND APPLIED SCIENCE, COLUMBIA UNIVERSITY: New York, NY

- Affiliate Faculty in Computer Science
- Affiliate Faculty in Electrical Engineering

JUNE 2023 – DEC 2025
JUNE 2024 – DEC 2025

EDUCATION

HARVARD UNIVERSITY: Cambridge, MA

- PhD Engineering Sciences: Electrical Engineering with a focus in Robotics; GPA: 3.9/4.0
▪ Dissertation: “GPU Acceleration for Real-time, Whole-Body, Nonlinear Model Predictive Control”
▪ Advisors: Vijay Janapa Reddi and Scott Kuindersma
- MEng in Engineering Sciences: Electrical Engineering with a focus in Robotics; GPA: 3.9/4.0
▪ Thesis: “Parallel and Constrained Differential Dynamic Programming for Model Predictive Control”
▪ Advisor: Scott Kuindersma

MAY 2022

MAY 2018

MASSACHUSETTS INSTITUTE OF TECHNOLOGY (MIT): Cambridge, MA

- Advanced Study Non-Degree Student; GPA: 5.0/5.0

JUNE 2016

HARVARD UNIVERSITY: Cambridge, MA

- B.A. Magna Cum Laude in Computer Science with a Minor in Economics; GPA: 3.9/4.0
▪ Thesis: “Hacking the White House: Election Fraud in the Digital Age”
▪ Advisor: Greg Morrisett

MAY 2013

AWARDS AND RECOGNITION

- Barnard College Teaching Excellence Award – Finalist MAY 2025
- Emily Gregory Award for Devotion to Teaching and Serving to the Students of Barnard College – Finalist APRIL 2024, 25
- Best Poster Award at the TC on Model Based Optimization for Robotics Virtual Poster Session JULY 2024
- Best Paper in Automation at ICRA 2024 MAY 2024
- Best Conference Paper at ICRA 2024 – Finalist APRIL 2024
- Best Student Paper at ICRA 2024 – Finalist APRIL 2024
- Best Poster Award at the IROS Workshop: Methods for Objective Comparison of Results in Intelligent Robotics Research OCT 2023
- Blue Ribbon TC Award given to the RAS TC on Model-Based Optimization for Robotics while Co-Chair SEPT 2023
- IEEE Micro Top Picks – Honorable Mention JAN 2022
- Harvard Faculty of Arts and Sciences Certificate in Undergraduate Mentoring APRIL 2021
- The Derek Bok Center Distinction in Teaching Award FALL 2017,18,19,20
- The Derek Bok Center Teaching Certificate DEC 2019
- Best Poster Award at the ICRA Workshop: Toward Online Optimal Control of Dynamic Robots MAY 2019
- National Science Foundation Graduate Research Fellowship (NSF GRFP) APRIL 2018

GRANTS

All award amounts represent my share of the grant if not otherwise specified.

ACTIVE AWARDS

- Lead PI for the [National Science Foundation \(NSF\)](https://www.nsf.gov/) 25-543 Computer and Information Science and Engineering: Future Computing Research (Future CoRe):
“Collaborative Research: SHF: Scalable Cross-Layer Co-Design for Stable Mixed-Precision Acceleration”
[Award #2621699: \\$405,886](#) 2027 – 29
- University PI (Lead-PI) for the [Toyota Research Institute \(TRI\)](https://www.toyota.com/research/) University 3.0:
“Real-Time Uncertainty-Aware Optimization for Intelligent Control”
[Award: \\$1,001,474](#) 2026 – 29
- Lead PI for the [National Science Foundation \(NSF\)](https://www.nsf.gov/) 24-581 Cyber Physical Systems (CPS):
“Collaborative Research: CPS: Medium: Safe Multi-Task Optimal Control for Drone Swarms: Theory to Deployment via Co-Design”
[Award #2535096: \\$498,819](#) 2026 – 29

- Co-PI for the IEEE Robotics and Automation Society Technical Education Programs (RAS-TEP): 2026 – 27
“2nd Optimization for Robotics Summer School”
Award: \$25,000 to support the costs of the school and to offer scholarships to participants from underrepresented groups
- Co-PI for the Dartmouth College University Seminars 2026 – 27
“Dartmouth Autonomy Series”
Award: \$6,000 to support travel and logistics for invited speakers
- Sole PI for the Burke Research Initiation Award 2025 – 31
“Optimizing Robotic Systems at All Scales”
Award: \$40,000
- Sole PI for the National Science Foundation (NSF) 22-632 Cyberinfrastructure for Sustained Scientific Innovation (CSSI): 2024 – 27
“Elements: RUI: Accessible GPU-Accelerated Edge Optimal Control Library and Benchmarks”
Award #2411369: \$590,548
 - Moved to Co-PI Subaward of \$352,863 for FY 26 and FY 27 after Transition to Dartmouth

PRIOR AWARDS

- University PI (Lead-PI) for the Toyota Research Institute (TRI) University 2.0: 2025 – 26
“Uncertainty-Aware Optimization for Intelligent Control”
Award: \$129,500
- Co-PI for the IEEE Robotics and Automation Society Technical Education Programs (RAS-TEP): 2024 – 25
“Optimization for Robotics Summer School”
Award: \$25,000 to support the costs of the school and to offer scholarships to participants from underrepresented groups
- Sole PI for the National Science Foundation (NSF) 23-576 Computer and Information Science and Engineering Research Initiation Initiative (CRII): 2023 – 25
“CRII: OAC: RUI: Real-Time, Mixed-Integer Model Predictive Control via Learned GPU-Acceleration”
Award #2246022: \$174,957
 - Additionally Awarded a Research Experiences for Undergraduates (REU) Supplement – \$24,000
 - Additionally Awarded a Career-Life Balance (CLB) Supplement – \$24,898
- Co-PI for the IEEE Computer Society Diversity and Inclusion Fund: 2022
“TinyML Outreach Workshop with the Navajo Nation”
Award: \$5,000 for stipends and hardware for workshop participants

PUBLICATIONS

Google Scholar: h-index=19; i10-index=28

Conferences are the primary publication venue in robotics and senior authors are typically listed last

PREPRINTS (UNDER REVIEW)

- (P1) G. Bravo-Palacios, J. Zhang, Z. Pestrnikov, **B. Plancher**, T. Lew, “TurboMPC: Fast, Scalable, and Differentiable Model Predictive Control on the GPU,” arXiv, June, 2026.
- (P2) Y. Jiang, R. Xing, L. Zhao, **B. Plancher**, M. Chen, D. Balkcom, “ASCII Art Turns LLMs into VLA Controllers,” March 2026.
- (P3) T. Desai, **B. Plancher**, R. I. Bahar, “Real-Time, Energy-Efficient, Sampling-Based Optimal Control via FPGA Acceleration,” arXiv, January 2026.
- (P4) S. Yang, T. Ohtsuka, **B. Plancher**, C. N. Jones, “Polynomial and Parallelizable Preconditioning for Block Tridiagonal Positive Definite Matrix,” arXiv, May 2025.

JOURNAL ARTICLES

- (J1) T. Lew, M. Greiff, J. Subosits, **B. Plancher**, “Solving Quadratic Programs with Slack Variables via ADMM without Increasing the Problem Size,” European Control Journal (EJC), July 2026.
 - Special Issue for the Best Papers from the European Control Conference (ECC), Reykjavík, Iceland, July 2026.
- (J2) J. Kent, E. Stefani, **B. Plancher**, “Robust Geospatial Coordination of Multi-Agent Communications Networks Under Attrition,” in IEEE Robotics and Automation Letters (RA-L), April 2026.
- (J3) A. Yilmaz, T. Bourgeat, L. Pentecost, **B. Plancher**, S. M. Neuman “RoboPrec: Enabling Reliable Embedded Computing for Robotics by Providing Accuracy Guarantees Across Mixed- Precision Datatypes,” in IEEE Robotics and Automation Letters (RA-L), December 2025.
- (J4) M. Sloane, E. Moss, S. Kennedy, M. Stewart, P. Warden, **B. Plancher**, V. Janapa Reddi, “Materiality and Risk in the Age of Pervasive AI Sensors,” in Nature Machine Intelligence, March 2025.
- (J5) B. Boroujerdian, H. Genc, S. Krishnan, P. Bardienus, B. Duisterhof, **B. Plancher**, K. Mansoorshahi, M. Almeida, A. Faust, V. Janapa Reddi. “The Role of Compute in Autonomous Aerial Vehicles.” in IEEE Transactions on Computers, July 2022.
- (J6) V. Janapa Reddi, **B. Plancher**, S. Kennedy, L. Moroney, P. Warden, A. Agarwal, C. Banbury, M. Banzi, M. Bennett, B. Brown, S. Chitlangia, R. Ghosal, S. Grafman, R. Jaeger, S. Krishnan, M. Lam, D. Leiker, C. Mann, M. Mazumder, D. Pajak, D. Ramaprasad, J. E. Smith, M. Stewart, D. Tingley, “Widening Access to Applied Machine Learning with TinyML,” in Harvard Data Science Review, January 2022.

- (J7) **B. Plancher**, S. M. Neuman, T. Bourgeat, S. Kuindersma, S. Devadas, V. Janapa Reddi, “Accelerating Robot Dynamics Gradients on a CPU, GPU, and FPGA,” in IEEE Robotics and Automation Letters (RA-L), January 2021.
- Additionally dual-accepted under the RAL+ICRA option for presentation at the IEEE International Conference on Robotics and Automation (ICRA), Xi’an China and Virtual, June 2021.

REFEREED CONFERENCE PROCEEDINGS

- (C1) R. Xing, S. Ree, **B. Plancher**, “MPC-Injection: Biasing Off-Policy Locomotion RL Toward Controller-Induced Behavior Basins,” in the Conference on Robot Learning (CoRL), Austin, TX, USA, November 2026.
- (C2) C. Yasutake, A. Liu, Z. Kingston, **B. Plancher**, “HJCD-IK: GPU-Accelerated Inverse Kinematics through Batched Hybrid Jacobian Coordinate Descent,” in the IEEE/RSJ International Conference on Intelligent Robots and Systems (iROS), Pittsburgh, PA, USA, October 2026.
- (C3) L. Zhang, G. Bravo-Palacios, **B. Plancher**, S. Grammatico, “Parallel Dynamic Programming for Conic Linear Quadratic Control,” in the IFAC World Congress, Busan, Republic of Korea, August 2026.
- (C4) I. Mahajan, J. Arrizabalaga, A. Grillo, F. Vega, J. Anderson, Z. Manchester, **B. Plancher**, “TinySDP: Real Time Semidefinite Optimization for Certifiable and Agile Edge Robotics,” in Robotics Science and Systems (RSS), Sydney, Australia, July 2026.
- (C5) A. Du, E. Adabag, G. Bravo-Palacios, **B. Plancher**, “GATO: GPU-Accelerated and Batched Trajectory Optimization for Scalable Edge Model Predictive Control,” in the IEEE International Conference on Robotics and Automation (ICRA), Vienna, Austria, June 2026.
- (C6) C. Sha, J. Jiang, A. Bhakta, K. Qiu, I. Mahajan, G. Bravo-Palacios, **B. Plancher**, “TAG-K: Tail-Averaged Greedy Kaczmarz for Computationally Efficient and Performant Online Inertial Parameter Estimation,” in the IEEE International Conference on Robotics and Automation (ICRA), Vienna, Austria, June 2026.
- (C7) C. Huang, P. Jadhav, **B. Plancher**, Z. Kingston, “pRRTC: GPU-Parallel RRT-Connect for Fast, Consistent, and Low-Cost Motion Planning,” in the IEEE International Conference on Robotics and Automation (ICRA), Vienna, Austria, June 2026.
- (C8) I. Mahajan, K. Nguyen, S. Schoedel, E. Nedumaran, M. Mata, **B. Plancher**, Z. Manchester, “Code Generation and Conic Constraints for Model-Predictive Control on Microcontrollers with Conic-TinyMPC,” in the IEEE International Conference on Robotics and Automation (ICRA), Vienna, Austria, June 2026.
- (C9) Z. Wu, **B. Plancher**, I. McInerney, M. Wang, K.-T. Cheng, “MPC Solver Hardware Generation Framework with Model-Specific Operation Fusion and Pruning,” in the International Conference on Field Programmable Technology (FPT), Shanghai, China, December 2025.
- (C10) I. Mahajan, **B. Plancher**, “Robust and Efficient Embedded Convex Optimization through First-Order Adaptive Caching,” in the IEEE/RSJ International Conference on Intelligent Robots and Systems (iROS), Hangzhou, China, October 2025.
- (C11) C. J. Carver, H. Schwartz, T. Itagaki, Z. Englhardt, K. Liu, M. Manik, C. Chang, V. Iyer, **B. Plancher**, X. Zhou, “Set Phasers to Stun: Beaming Power and Control to Mobile Microrobots with Laser Light,” in the IEEE/RSJ International Conference on Intelligent Robots and Systems (iROS), Hangzhou, China, October 2025.
- (C12) E. Selman-Housein, **B. Plancher**, “Improving the Representation of Undergraduate Women in Cybersecurity: A Literature Review,” ACM Technical Symposium on Computer Science Education (SIGCSE-TS), Pittsburg, PA, USA, February 2025.
- (C13) S. M. Neuman, **B. Plancher**, V. Janapa Reddi, “Invited: The Magnificent Seven Challenges and Opportunities in Domain-Specific Accelerator Design for Autonomous Systems,” ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, USA, June 2024.
- (C14) K. Nguyen, S. Schoedel, A. Alavilli, **B. Plancher**, Z. Manchester, “TinyMPC: Model-Predictive Control on Resource-Constrained Microcontrollers,” in the IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, May 2024.
- Best Paper in Automation at IEEE ICRA 2024
 - Finalist for Best Conference Paper at IEEE ICRA 2024
 - Finalist for Best Student Paper at IEEE ICRA 2024
- (C15) E. Adabag, M. Atal, W. Gerard, **B. Plancher**, “MPCGPU: Real-Time Nonlinear Model Predictive Control through Preconditioned Conjugate Gradient on the GPU,” in the IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, May 2024.
- Best Poster Award at the 2024 IEEE TC on Model Based Optimization for Robotics Virtual Poster Session
- (C16) X. Bu and **B. Plancher**, “Symmetric Stair Preconditioning of Linear Systems for Parallel Trajectory Optimization,” in the IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, May 2024.
- (C17) L. Grossman and **B. Plancher**, “Differentially Encoded Observation Spaces for Perceptive Reinforcement Learning,” in the IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, May 2024.
- (C18) V. Mayoral-Vilches, J. Jabbour, Y. Hsiao, Z. Wan, A. Martínez-Fariña, M. Crespo-Álvarez, M. Stewart, J. Reina-Muñoz, P. Nagras, G. Vikhe, M. Bakhshalipour, M. Pinzger, S. Rass, S. Panigrahi, G. Corradi, N. Roy, P. B. Gibbons, S. M. Neuman, **B. Plancher**, V. Janapa Reddi, “RobotPerf: An Open-Source, Vendor-Agnostic, Benchmarking Suite for Evaluating Robotics Computing System Performance,” in the IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, May 2024.
- Best Poster Award for preliminary results at the Workshop on Methods for Objective Comparison of Results in Intelligent Robotics Research at the 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (iROS)

- (C19) **B. Plancher**, M. Zennaro, M. Rovai, V. Janapa Reddi, S. Buttrich, J. Ellis, N. Goveas, L. Kazimierski, J. Lopez Sotelo, M. Lukic, D. Mendez, R. Nordin, A. Oliva Trevisan, M. Pavan, M. Roveri, M. Rub, J. Tum, M. Verhelst, S. Abdeljabar, S. Adebayo, T. Amberg, H. Aworinde, J. Bagur, G. Barrett, N. Benamar, B. Chaudhari, R. Criollo, D. Cuartielles, J. A. Ferreira Filho, S. Gizaw, E. Gousev, A. Grande, S. Hymel, P. Ing, P. Manandhar, P. Manzoni, B. Murmann, E. Pan, R. Paskauskas, E. Pietrosemoli, T. Pimenta, “TinyML4D: Scaling Embedded Machine Learning Education in the Developing World,” AAAI 2024 Spring Symposium on Increasing Diversity in AI Education and Research, San Francisco, CA, USA, March 2024.
- Invited to be Presented as an Extended Session at the Symposium [\[Link\]](#).
- (C20) S. M. Neuman, R. Ghosal, T. Bourgeat, **B. Plancher**, V. Janapa Reddi, “RoboShape: Using Topology Patterns to Scalably and Flexibly Deploy Accelerators Across Robots,” in the International Symposium on Computer Architecture (ISCA), Orlando, FL, USA, June 2023.
- (C21) L. Grossman and **B. Plancher**, “Just Round: Quantized Observation Spaces Enable Memory Efficient Learning of Dynamic Locomotion,” in the IEEE International Conference on Robotics and Automation (ICRA), London, UK, May 2023.
- (C22) J. Blanchard, J. R. Hott, V. Berry, R. Carroll, B. Edmison, R. Glassey, O. Karnalim, **B. Plancher**, S. Russell, “Stop Reinventing the Wheel! Promoting Community Software in Computing Education,” in the 2022 Working Group Reports on Innovation and Technology in Computer Science Education (ITiCSE-WGR), Dublin, Ireland, December 2022.
- (C23) V. Mayoral-Vilches, S. M. Neuman, **B. Plancher**, V. Janapa Reddi, “RobotCore: An Open Architecture for Hardware Acceleration in ROS 2,” in the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Kyoto, Japan, October 2022.
- (C24) S. M. Neuman, **B. Plancher**, B. P. Duisterhof, S. Krishnan, C. Banbury, M. Mazumder, S. Prakash, J. Jabbour, A. Faust, C.H.E. de Croon, V. Janapa Reddi, “Tiny Robot Learning: Challenges and Directions for Machine Learning in Resource-Constrained Robots,” in the IEEE International Conference on Artificial Intelligence Circuits and Systems (AICAS), Incheon, Republic of Korea, June 2022.
- (C25) **B. Plancher**, S. M. Neuman, R. Ghosal, S. Kuindersma, V. Janapa Reddi, “GRiD: GPU-Accelerated Rigid Body Dynamics with Analytical Gradients,” IEEE International Conference on Robotics and Automation (ICRA), Philadelphia PA, May 2022.
- (C26) B. Boroujerdian, R. Ghosal, J. Cruz, **B. Plancher**, V. Janapa Reddi, “RoboRun: A Robot Runtime to Exploit Spatial Heterogeneity,” in the Design Automation Conference (DAC), Virtual, December 2021.
- (C27) S. M. Neuman, **B. Plancher**, T. Bourgeat, T. Tambe, S. Devadas, V. Janapa Reddi, “Robomorphic Computing: A Design Methodology for Domain-Specific Accelerators Parameterized by Robot Morphology,” in the ACM International Conference on Architecture Support for Programming Languages and Operating Systems (ASPLOS), Virtual, April 2021.
- IEEE Micro Top Picks 2022 – Honorable Mention
- (C28) **B. Plancher**, C. Brumaar, I. Brumar, L. Pentecost, S. Rama, D. Brooks, “Application of Approximate Matrix Multiplication to Neural Networks and Distributed SLAM,” in the IEEE High Performance Extreme Computing Conference (HPEC), Waltham, MA, September 2019.
- (C29) **B. Plancher** and S. Kuindersma, “A Performance Analysis of Parallel Differential Dynamic Programming on a GPU,” in the Workshop on the Algorithmic Foundations in Robotics (WAFR), Merida, Mexico, December 2018.
- (C30) **B. Plancher**, Z. Manchester, S. Kuindersma, “Constrained Unscented Dynamic Programming,” in the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Vancouver, Canada, September 2017.
- (C31) S. Karaman, A. Anders, M. Boulet, J. Connor, K. Gregson, W. Guerra, O. Guldner, M. Mohamoud, **B. Plancher**, R. Shin, J. Vivilecchia, “Project-based, collaborative, algorithmic robotics for high school students: Programming self-driving race cars at MIT,” in the IEEE Integrated STEM Education Conference (ISEC), Princeton, NJ, March, 2017.

REFEREED MAGAZINE ARTICLES AND COLUMNS

- (M1) M. Stewart, Y. Zhang, P. Warden, Y. Omri, S. Prakash, J. Huckelberry, J. Santos, S. Hymel, B. Brown, J. MacArthur, N. Jeffries, E. Moss, M. Sloane, **B. Plancher**, V. Janapa Reddi, “Datasheets for Machine Learning Sensors,” AI Magazine, Vol. 47, January 2026.
- (M2) K. Chatzilygeroudis, S. Kumar, **B. Plancher**, Z. Manchester, E. Mingo Hoffman, P. Wensing, “2025 Summer School on Optimization for Robotics [Education],” IEEE Robotics and Automation Magazine (RAM), December 2025.
- (M3) E. Fields, C. Ho, M. J. Kim, Z. Wu, **B. Plancher**, “Underrepresentation of Women in Robotics Research [Women in Engineering],” IEEE Robotics and Automation Magazine (RAM), March 2024.
- (M4) S. Prakash, M. Stewart, C. Banbury, M. Mazumder, P. Warden, **B. Plancher**, V. Janapa Reddi, “Is TinyML Sustainable? Assessing the Environmental Impacts of Machine Learning on Microcontrollers,” Communications of the ACM (CACM), November 2023.
- (M5) P. Warden, M. Stewart, **B. Plancher**, S. Katti, V. Janapa Reddi, “Machine Learning Sensors: A Design Paradigm for the Future of Intelligent Sensors,” Communications of the ACM (CACM), November 2023.

REFEREED WORKSHOP PUBLICATIONS AND POSTERS

- (W1) I. Mahajan, B. Plancher, “LIMO: Learning Inside Model Predictive Optimization for Recoverability-Aware Robot Control,” Workshop on Failure Recovery at RSS 2026.

- (W2) A. Grillo, **B. Plancher**, “End-to-End Hardware-Algorithm Co-Design for High-Rate FPGA-Accelerated Model Predictive Control on Tiny Drones,” Frontiers of Optimization (FOR) Workshop and Workshop on Robotics Acceleration with Computing Hardware and Systems. (RoboARCH) at ICRA 2026.
- (W3) C. Huang, **B. Plancher**, Z. Kingston, “pAORRTC: GPU-Parallel Almost-Surely Asymptotically Optimal Planning,” Frontiers of Optimization (FOR) Workshop and Workshop on Robotics Acceleration with Computing Hardware and Systems. (RoboARCH) at ICRA 2026.
- (W4) G.-, J. Zhang, Z. Pestrikov, **B. Plancher**, T. Lew, “Differentiable and Constrained Model Predictive Control on the GPU,” Frontiers of Optimization (FOR) Workshop and Workshop on Robotics Acceleration with Computing Hardware and Systems. (RoboARCH) at ICRA 2026.
- (W5) A. Yilmaz, L. Pentecost, T. Bourgeat, **B. Plancher**, S. Neuman, “Provably Accurate Fixed-Point Arithmetic for Robotics Accelerators,” Workshop on Robotics Acceleration with Computing Hardware and Systems. (RoboARCH) at ICRA 2026.
- (W6) I. Mahajan, J. Arrizabalaga, A. Grillo, F. Vega, J. Anderson, Z. Manchester, **B. Plancher**. “TinySDP: Real Time Semidefinite Optimization for Certifiable and Agile Edge Robotics,” Workshop on Robotics Acceleration with Computing Hardware and Systems. (RoboARCH) at ICRA 2026.
- (W7) T. Desai, **B. Plancher**, R. I. Bahar “Dataflow MPPI: Hardware-Software Co-Design for FPGA Acceleration of Sampling-Based Optimal Control,” Workshop on Robotics Acceleration with Computing Hardware and Systems. (RoboARCH) at ICRA 2026.
- (W8) T. Desai, **B. Plancher**, R. I. Bahar. “Energy Efficient, Real-Time Robotic Path Planning through FPGA Acceleration,” Late Breaking Results Session at the Design Automation Conference (DAC), San Francisco, CA, USA, June 2025.
- (W9) E. Adabag, X. Bu, K. Nguyen, S. Schoedel, A. Alavilli, M. Atal, W. Gerard, E. Nedurmaran, Z. Manchester, **B. Plancher**. “Optimizing at All Scales: Edge (Non)linear Model Predictive Control from MCUs to GPUs,” Frontiers of Optimization for Robotics Workshop at the Robotics Science and Systems (RSS), Delft, Netherlands, July 2024.
 - Additionally Invited for a Spotlight Talk at the Workshop
- (W10) **B. Plancher**, “Parallel Optimization for Robotics: An Undergraduate Introduction to GPU Parallel Programming and Numerical Optimization Research,” IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW), San Francisco, CA, USA, May 2024.
- (W11) W. Xie, **B. Plancher**, “Can Large Language Models Reduce the Barriers to Entry for High School Robotics?” Robots for Learning Workshop at the IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN), Busan, Republic of Korea and Remote, August 2023.
- (W12) **B. Plancher**, “Tiny Robot Learning: Expanding Access to Edge ML as a Step Toward Accessible Robotics,” Lowering Barriers for Robotics Research Workshop at the Robotics Science and Systems (RSS), Daegu, Republic of Korea and Remote, July 2023.
- (W13) M. Zennaro, **B. Plancher**, V. Janapa Reddi, “Bridging the Digital Divide: The Promising Impact of TinyML for Developing Countries,” at the UN 8th Multi-stakeholder Forum on Science, Technology and Innovation for the Sustainable Development Goals, New York, NY, USA and Remote, May 2023.
- (W14) S. Kennedy and **B. Plancher**, “Voice Interfaces, Gender, and Race: An Intersectional Analysis,” in the Gendering Robots (GenR) Workshop at the IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), Naples, Italy, August 2022.
- (W15) J. Blanchard, J. R. Hott, V. Berry, R. Carroll, B. Edmison, R. Glassey, O. Karnalim, **B. Plancher**, S. Russell, “Leveraging Community Software in CS Education to Avoid Reinventing the Wheel,” in the ACM Conference on Innovation and Technology in Computer Science Education (ITiCSE), Dublin, Ireland, July 2022.
- (W16) J. Jabbour, S. M. Neuman, M. Mazumder, C. Banbury, S. Prakash, **B. Plancher**, and V. Janapa Reddi, “Closing the Sim-to-Real Gap for Ultra-Low-Cost, Resource-Constrained, Quadrupe Robot Platforms,” in the Closing the Sim2Real Gap workshop at Robotics Science and Systems (RSS), New York, New York, June 2022.
- (W17) M. Zennaro, **B. Plancher**, and V. Janapa Reddi, “TinyML: Applied AI for Development,” at the UN 7th Multi-stakeholder Forum on Science, Technology and Innovation for the Sustainable Development Goals. New York, NY, USA and Remote. May 2022.
- (W18) **B. Plancher** and V. Janapa Reddi, “The Tiny Machine Learning Open Education Initiative (TinyMLedu),” ACM Technical Symposium on Computer Science Education (SIGCSE), Providence, Rhode Island, March 2022.
- (W19) **B. Plancher** and S. Kuindersma, “Realtime Model Predictive Control using Parallel DDP on a GPU,” in the workshop Toward Online Optimal Control of Dynamic Robots at the International Conference on Robotics and Automation (ICRA), Montreal, Canada, May 2019.
 - Winner of the Best Poster Award at the Workshop

OTHER PUBLICATIONS AND POSTERS

- (O1) S. Yang, T. Ohtsuka, **B. Plancher**, C. N. Jones, “Polynomial and Parallelizable Preconditioning for Block Tridiagonal Positive Definite Matrix,” Workshop of the GAMM Activity Group on Applied and Numerical Linear Algebra (ANLA), Bologna, Italy, October, 2025.
- (O2) J. Kent, E. Stefani, **B. Plancher**, “Robust Geospatial Coordination of Multi-Agent Communications Networks Under Attrition,” Northeast Robotics Colloquium (NERC), Ithaca, New York, October, 2025.
- (O3) J. Jabbour, J. Wang, **B. Plancher**, “Receding Diffusion: Domain Informed, Efficient Diffusion Path Planning for Dynamic Environments,” Northeast Robotics Colloquium (NERC), Amherst, Massachusetts, September 2024.

- (O4) A. E. Yilmaz, T. Bourgeat, **B. Plancher**, S. M. Neuman, “Accelerating Rigid Body Dynamics through Variable Fixed-Precision with Provable Error-Bounds,” Northeast Robotics Colloquium (NERC), Amherst, Massachusetts, September 2024.
- (O5) K. Awotwi, Z. Pestrikov, A. Sharma, D. Tuchman, **B. Plancher**, “Towards a Comprehensive GPU-Accelerated Rigid Body Dynamics Library: Extending GRiD,” Northeast Robotics Colloquium (NERC), Amherst, Massachusetts, September 2024.
- (O6) A. Du, **B. Plancher**, “Towards Real-Time Batched Trajectory Optimization with MPCGPU,” Northeast Robotics Colloquium (NERC), Amherst, Massachusetts, September 2024.
- (O7) M. Zennaro, **B. Plancher**, M. Stewart, V. Janapa Reddi, “AI in the developing world: how ‘tiny machine learning’ can have a big impact,” The Conversation, February 2024.
- (O8) E. Adabag, M. Atal, W. Gerard, **B. Plancher**, “Accelerating Nonlinear MPC with Warm Started Iterative Solvers on GPUs,” the IEEE-RAS Technical Committee on Model-Based Optimization for Robotics Virtual Poster Session, Remote, July 2023.
- (O9) C. Ho and **B. Plancher**, “Gender Diversity in Robotics Research,” Northeast Robotics Colloquium (NERC), Lowell, Massachusetts, October 2022.
- (O10) P. Warden, M. Stewart, **B. Plancher**, C. Banbury, S. Prakash, E. Chen, Z. Asgar, S. Katti, and V. Janapa Reddi, “Machine Learning Sensors,” arXiv preprint, June 2022.

TEACHING

Dartmouth College: Hanover, NH – *Instructor of Record*

FALL 2025 – PRESENT

- **COSC 2/81: Principles of Robot Design and Programming** (Fall 2026)
 - Re-designed and adapted lecture slides, problem sets, programming assignments, and exams for the 40 student course.
 - Maintained overall structure, and project-based components
- **COSC 1/69.23: Parallel Optimization for Robotics** (Fall 2025)
 - Adapted COMS BC 3159 from Barnard to Dartmouth to adjust to quarter system length courses
 - Maintained overall structure, and project-based components

BARNARD COLLEGE, COLUMBIA UNIVERSITY: New York, NY – *Instructor of Record*

FALL 2022 – SPRING 2025

- **CSEE W3827: Fundamentals of Computer Systems** (Spring 2025)
 - Taught the 330-student, undergraduate course (in two back-to-back ~165 student lectures) which is required for the Computer Science, Electrical Engineering, and Computer Engineering majors.
 - Re-designed and adapted lecture slides, problem sets, programming assignments, and exams.
 - Developed a new optional recitation section in collaboration with select teaching assistants to address feedback from past course offerings on a need for additional worked examples and to tie weekly content to cross-cutting themes.
 - Managed a team of 15 teaching assistants to ensure that office hours were held, exams and homework assignments were graded, and student questions on the online forum were answered.
- **COMS BC 3159: Parallel Optimization for Robotics** (Spring 2023, 24, Fall 24)
 - Designed a new 50 student course (increased from 30 students in Spring 23) at the intersection of robotics and both numerical optimization and computer architecture / systems exploring the use of parallel programming on CPUs and GPUs to accelerate numerical optimization algorithms through the lens of robot motion planning and control
 - Designed and gave lectures on computer system design, parallel programming on the CPU and GPU using (CUDA) C++, (nonlinear) (trajectory) optimization, and numerical optimal control
 - Developed written assignments and coding assignments in both python and (CUDA) C++ covering the core course topics as well as a hands-on, team-based, final project to enable students to integrate and apply their learnings from the semester and practice presenting and technical writing. Student project highlights include:
 - “Preconditioned Conjugate Gradient on the GPU” in Spring 2023 which led to an ICRA 2024 Publication (C15)
 - “Accelerating RRT on the GPU” in Spring 2024 which led to an ICRA 2026 Publication (C7)
 - “Accelerating Inverse Kinematics on the GPU” in Spring 2024 which led to an IROS 2026 Publication (C2)
 - Published a Refereed Workshop paper on the design and pilot of the course at EduPar-24 (W10)
- **COMS BC 3449: Applied Computing – Research and Industry Perspectives** (Spring 2023, 24, Fall 24)
 - Designed a new 15 student course (reduced from 30 to increase attention to each student’s unique project) providing a unifying structure to support hands-on, computational student projects ranging from academic research to industry collaborations to independent passion projects (course ran as COMS BC 3997: New Directions in Computing rotating topics course in Spring 23, 24 and called Projects in Computer Science in Spring 23). Student project highlights include:
 - A multiple pivot LLM project in Spring 2024 which led to a \$250k pre-seed investment
 - Two continuations of COMS BC 3159 projects which led to ICRA 2026 (C7) and IROS 2026 (C2) Publications
 - Designed lectures on: performance engineering, cyber security, web development, and technical writing and presentation
 - Recruited guest speakers from academia and industry to present on topics including: responsible AI, startup engineering, accessible user interface design, working with legacy software systems, and the business of technology
 - Mentored student projects with one-on-one writing and presenting support, high level direction, and technical debugging
- **COMS BC 3997 - F22: New Directions in Computing: Introduction to Robotics Engineering** (Fall 2022)
 - Designed a new 20 student course providing a hands-on introduction to computational robotics for computer scientists
 - Designed and gave lectures covering algorithms for perception, mapping and localization, planning, control, and learning

- Developed written assignments and coding assignments in python as well as a hands-on final project using physical robot hardware to connect the algorithms learned in class into the physical world and reinforce their tradeoffs

edX Inc: Cambridge, MA – *Teaching Staff Lead*

SUMMER 2020 – 2022

- **HarvardX Professional Certificate in Tiny Machine Learning (TinyML) MOOC** [[Course 1-3](#) [Link](#) [Course 4](#)]
 - Co-designed a free, hands-on, project-based professional certificate taught through three 6-week courses (and an optional fourth course) on the EdX platform covering the emerging field of Tiny Machine Learning (deploying machine learning onto microcontrollers for machine learning at the edge) with the aim of democratizing access to this developing field
 - Over 100,000 students from over 190 countries enrolled as of July 2024 since the four courses launched in September 2020, December 2020, February 2021, and March 2022 respectively
 - Served as the laboratory instructor both co-designing hands-on exercises as well as recording video walkthroughs
 - Managed the 10-person course staff to ensure that content was created, reviewed, and produced in a timely manner
 - Led and managed external relations for the course team coordinating with edX, Google, and Arduino
 - Co-designed course materials including video lectures, readings, code walkthroughs, assessments, and discussion forums
 - Released all course materials open-source to enable global adaptation for further access to TinyML education [[link](#)]

HARVARD UNIVERSITY: Cambridge, MA – *Head Teaching Fellow (Head TA)*

FALL 2017-20

- **CS 249r - Special Topics in Edge Computing - Autonomous Machines** (Fall 2019) and **Tiny Machine Learning** (Fall 2020)
 - Co-designed 40-50 student courses at the intersection of artificial intelligence (robotics and machine learning) and computer architecture / embedded systems
 - Designed and gave lectures for the introduction to robotics and introduction to machine learning sections of the courses
 - Co-developed hands-on project-based assignments (e.g., training TinyML models with Google Colab and deploying on Arduinos) and course infrastructure/tools (e.g., online paper discussion forum)
 - Mentored student teams pursuing research-based final projects
- **CS 182: Introduction to Artificial Intelligence** (Fall 2017-2018)
 - Managed a team of 11 teaching fellows supporting the 150 student course to ensure sections and office hours were held, exams and homework assignments were graded, and student questions on the online forum were answered.
 - Designed and gave lectures: “Introduction to Robotics and Path Planning I/II” and the related assignment/exam questions
 - Co-Designed and gave a new set of weekly recitations that mapped the weekly course content to cross-cutting themes
 - Co-developed course coding and written (theoretical) assignments, and course infrastructure/tools (e.g., autograders)
 - Mentored student teams pursuing research-based final projects

MIT / HARVARD UNIVERSITY: Cambridge, MA – *Teaching Assistant*

FALL 2017,18,19,21

- Harvard’s 15 student section of MIT’s **MAS.863: How to Make Almost Anything**
 - Held office hours, led introductory sessions for course tools, aided students in lab work, machine usage, and project design
 - Gave recitations: “Introduction to Embedded Programming”, “Introduction to Electronics Fabrication and Design”

MIT LINCOLN LABORATORIES BEAVER WORKS: Cambridge, MA – *Associate Instructor*

SUMMER 2016,17,18,19

- **RACECAR Mini Grand Prix Challenge:** a hands-on, intensive, residential, project-based, free to attend, 4-week program for high school students interested in studying STEM
 - Worked with 9-12 teams of 4-6 students to teach programming concepts and robotic algorithm design through the completion of fast autonomous navigation tasks using 1/10 scale racecars using Python/ROS
 - Co-designed weekly challenges to ensure all teams developed the technical skills needed for the final race
 - Co-designed and co-built the final race track spanning an entire ice hockey rink

ADVISING

POSTDOCTORAL RESEARCH ADVISOR

- Roland Schwan (Dartmouth College): JUNE 2025 – PRESENT
Real-Time Uncertainty-Aware Optimization for Intelligent Control
- Gabriel Bravo-Palacios (Barnard College, Dartmouth College): FEB 2025 – PRESENT
Open-Source, Edge, GPU-Accelerated Optimal Control Libraries and Benchmarks

PHD ADVISOR

- Chih Huang (Dartmouth College) SEPT 2026 – PRESENT
Accelerated Sample-Based Motion Planning
- Aristeidis Androutsopoulos (Dartmouth College) SEPT 2026 – PRESENT
Next Generation Edge Control
- Roy Xing (Dartmouth College - NSF GRFP Fellow) SEPT 2025 – PRESENT
Accelerated Hybrid Control: From Learning to Optimization
- Jianghan Zhang (Dartmouth College) SEPT 2025 – PRESENT
Next Generation Optimization and Learning for Robotics
- Jonathan S. Kent (Columbia University): SEPT 2024 – PRESENT
Autonomous UxV Swarm Control and Communication
(Note: Co-Advised with Junfeng Yang of Columbia University following Transition to Dartmouth in Fall 2025)

MASTERS THESIS ADVISOR

- Ishaan Mahajan (Columbia University): SEPT 2024 – MAY 2026
Solvers at the Edge: Embedded Optimization for Safe and Intelligent Robotics
Next Role: PhD @ Carnegie Mellon University
- Andrea Grillo (École Polytechnique Fédérale de Lausanne - EPFL) SEPT 2025 – MAY 2026
Hardware-Algorithm Co-Design for Real-Time Linear Model Predictive Control
(Note: Co-Advised with Colin Jones of EPFL)

UNDERGRADUATE THESIS ADVISOR

- Rosh Ho (Columbia University): SEPT 2024 – MAY 2025
Achieving Dexterous Manipulation via Vision for Stoke Patients
- Khushi Tyagi (Barnard College, Columbia University): Distinction SEPT 2023 – MAY 2024
Fall Detection for the Elderly with Embedded Systems
Next Role: MPH @ Columbia University
- Dhilan Ramaprasad (Harvard University): Highest Honors SEPT 2021 – MAY 2022
Teaching Embedded Systems Programming
Next Role: High School STEM Teacher
- Lev Jacob Grossman (Harvard University): Highest Honors SEPT 2019 – MAY 2020
Reinforcement Learning to Enable Robust Robotic Model Predictive Control
Next Role: Machine Learning/Software Engineer @ Berkshire Grey
- John Alex Keszler (Harvard University): Honors SEPT 2018 – MAY 2019
FPGA Acceleration of Motion Planning Algorithms for Robotics Applications
Next Role: MS @ Massachusetts Institute of Technology

PHD THESIS COMMITTEE

- Derin Ozturk (Cornell University) AUG 2025 – PRESENT
Hardware Acceleration for Insect Scale Robotics
- Se Hwan Jeon (Massachusetts Institute of Technology - MIT) NOV 2025 – MAY 2026
Structuring Reinforcement Learning for Legged Robots with GPU-Parallelized MPC

MASTERS THESIS COMMITTEE

- Basel Nitham Hindi (Columbia University) DEC 2023
Computer Vision-Powered Applications for Interpreting and Interacting with Movement

PHD MILESTONE COMMITTEE MEMBER

- Yitao Jiang (Dartmouth College) – Research Proposal Exam MAY 2026
- Dhawal Sirkonda (Dartmouth College) – Research Proposal Exam FEB 2026

PRE-DOCTORAL RESEARCH ADVISOR

- Next Role PhD: Emre Adabag (University of Michigan – CRA Undergraduate Research Award Finalist),
Highlights: William Xie (University of Colorado, Boulder – NSF GRFP Fellow), Moises Mata (University of Southern California – NSF GRFP Honorable Mention), Chih Huang (Dartmouth College), Danelle Tuchman (Tufts University), Han Lin (University of North Carolina, Chapel Hill)
- MS: Patarada Yontrarak, Kimiya Shahamat, Susannah Abrams, En Kai “Gino” Zhang, Jiayi Wang, Seojin “Heidi” Yoon (Columbia University), Seyoung Ree (Harvard University), Cael Yasutake (University of Pennsylvania), Abhinav Sharma (University of Massachusetts at Amherst)
- Industry: William Gerard (Senior Machine Learning Engineer, Bain & Company – NSF GRFP Honorable Mention), Eric Feng (ML Compiler Engineer, AMD), Miloni Atal (DevTech Engineer, NVIDIA), Yana Botvinnik (C++ Software Engineer, Splunk), Luci Feinberg (Software Engineer, Apple), Catherine Ji (Associate Product Manager, Google), Chau “Chloe” Nguyen (Software Engineer, Wayfair), Brennan McManus, Nick Thevenin (Software Engineering, Microsoft), Barkha Seth, Antonio Aranda (Software Development Engineering, Amazon), Srishti Srivastava (Software Engineer, Google), Claire Dawson (Engineer, Gecko Robotics), Alexander Du (Full Stack Robotics Software Engineer, Sunday Robotics), Nadim Kanazi (AI Research Engineer, Accenture)
- National Science Foundation Research Experiences for Undergraduates (NSF REU)
 - Seyoung Ree, Patarada Yontrarak: *GPU Acceleration of Optimal Control Solvers* SUMMER 2025
 - Alexander Du, Alice Lin: *GPU Acceleration of Optimal Control Solvers* SUMMER 2024
 - Columbia College Summer Funding Program, Columbia Engineering Summer Internship Program, Columbia School of General Studies Research Initiative for Summer Exploration, Columbia University Summer Undergraduate Research Experience (SURE) Program, and Egleston Scholars Summer Internship Program
 - Kai Zhang: *Constrained Parallel Differential Dynamic Programming* SUMMER 2025
 - Chih Huang: *Benchmarking and Acceleration of Sample-Based Motion Planning*
 - Danelle Tuchman: *GPU Acceleration of Rigid Body Dynamics* SUMMER 2024
 - Vidushi Jindal: *Embedded Perception, Mapping, and Localization*

- Kevin Han: *Federated Learning at the Edge for UAV Swarms*
- Alice Diakova: *Embedded Perception, Mapping, and Localization* SUMMER 2023
- Ena Selman-Housein: *Accessibility, Retention, & Diversity in Introductory CS and Cybersecurity*
- Jonathan Nalikka: *Constrained Parallel Differential Dynamic Programming*
- Barnard College Summer Research Institute (SRI)
 - Nicole Ineza: *Constrained Parallel Differential Dynamic Programming* SUMMER 2025
 - Aliya Tang: *Embedded Perception, Mapping, and Localization* SUMMER 2024
 - Nicole Ineza, Khadija Siddiqui, Annie Wang, Catherine Zhao: *Constrained Parallel Differential Dynamic Programming*
 - Yana Botvinnik, Xian Jiang: *GPU Acceleration of Linear System Solvers* SUMMER 2023
 - Claudia Lihar, Anagha Ram: *Constrained Parallel Differential Dynamic Programming*
 - Allyce Chung, Merrick Wolfley, Julianna Yu: *Accessibility, Retention, & Diversity in Introductory CS and Cybersecurity*
 - Emily Burnett, Neasha Mittal, Chau “Chloe” Nguyen, Kimiya Shahamat: *GPU Acceleration of Rigid Body Dynamics*
 - Aliya Tang, Elvina Wibisono, Britney Aparicio, Tramy Dong: *Embedded Perception, Mapping, and Localization*
 - Susannah Abrams: *GPU Acceleration of Rigid Body Dynamics* SUMMER 2022
 - Tara Bogavelli: *Sim-to-Real Reinforcement Learning for Low-Cost Walking Robots*
 - Chloe Ho: *Gender Diversity in Robotics Research*
- Other Mentored Independent and Group Research Projects
 - (Post-Masters) Patarada Yontrarak, (Masters) Cael Yasutake, (Post-Baccalaureate) Moises Mata, Chih Huang, Catherine Zhao (Undergraduate) Sean Mabli, Kai Zhang SUMMER 2026
 - (Masters) Seyoung Ree, Patarada Yontrarak, Cael Yasutake, (Undergraduate) Sean Mabli, Moises Mata, Kai Zhang, Chih Huang, Catherine Zhao, SPRING 2026
 - (Masters) Seyoung Ree, Patarada Yontrarak, Cael Yasutake, (Undergraduate) Sean Mabli, Zachary Pestrikov, Moises Mata, Kai Zhang, Chih Huang, Catherine Zhao, Alexander Du, Shuo Sha WINTER 2026
 - (Masters) Seyoung Ree, Patarada Yontrarak, Cael Yasutake, (Undergraduate) Sean Mabli, Zachary Pestrikov, Moises Mata, Kai Zhang, Chih Huang, Catherine Zhao, Alexander Du, Shuo Sha FALL 2025
 - (Post-Masters) Kwamena Awotwi, (Masters) Roy Xin, (Post-Baccalaureate) Cael Yasutake, (Undergraduate) Zachary Pestrikov, Alexander Du SUMMER 2025
 - (Post-Masters) Kwamena Awotwi, (Masters) Brennan McManus, Abhinav Sharma, Luci Feinberg, Venkat Bitra, (Post-Baccalaureate) Jiayi Wang, Elvina Wibisono, (Undergraduate) Seyoung Ree, Charles Chen, Brandon Pae, Moises Mata, Yahya ElGawady, Skyla Cui, Kai Zhang, Patarada Yontrarak, Anupam Bhakta, Justin Jiang, Kevin Qiu, Shuo Sha, Chih Huang, Cael Yasutake, Merrick Wolfley, Zachary Pestrikov, Emre Adabag, Alexander Du, Shobini Iyer, Danelle Tuchman, Aliya Tang SPRING 2025
 - (Post-Masters) Kwamena Awotwi, (Masters) Nick Thevinin, Nadim Kanazi, Claire Dawson, Venkat Bitra, Brenan McManus, Nagavasavi Jeepalyam, Abhinav Sharma, (Post-Baccalaureate) Yana Botvinnik, Jiayi Wang, (Undergraduate) En Kai “Gino” Zhang, Danelle Tuchman, Zachary Pestrikov, Anupam Bhakta, Justin Jiang, Kevin Qiu, Shuo Sha, Charles Chen, Alexander Du, Seyoung Ree, Louis Zheng, Nicole Ineza, Catherine Zhao, Chih Huang, Cael Yasutake, Aidan Eichman, Yahya ElGawady, Skyla Cui, Annie Wang, Nicole Ineza, Khadija Siddiqui, Catherine Zhao, Alice Lin, Alexander Du, Seojin “Heidi” Yoon, Louis Zheng, Aliya Tang, Elvina Wibisono, Shobini Iyer FALL 2024
 - (Masters) Eric Feng, Nadim Kanazi, Kwamena Awotwi, Abhinav Sharma, (Post-Baccalaureate) Yana Botvinnik, (Undergraduate) Zachary Pestrikov, Merrick Wolfley SUMMER 2024
 - (Masters) Eric Feng, William Gerard, Nagavasavi Jeepalyam, Yishen Zheng, Kwamena Awotwi, Naren Loganathan, Abhinav Sharma, Nicholas Thevenin, Brennan McManus, (Undergraduate) Yana Botvinnik, Seojin “Heidi” Yoon, Justin Haddad, Anupam Bhakta, Justin Jiang, Kevin Qiu, Shuo Sha, Carly Kiang, Chau “Chloe” Nguyen, Catherine Ji, Kimiya Shahamat, Danelle Tuchman, Britney Aparicio, Barkha Seth, Aliya Tang, Elvina Wibisono, Luci Feinberg, Shobini Iyer, Jiayi Wang, Chloe Ho, Merrick Wolfley, Emre Adabag, Alice Lin, Mikul Saravanan, Seyoung Ree SPRING 2024
 - (Masters) William Gerard, Abhinav Sharma, (Undergraduate) Carly Kiang, Chau “Chloe” Nguyen, Catherine Ji, Kimiya Shahamat, Danelle Tuchman, Britney Aparicio, Alice Diakova, Tramy Dong, Barkha Seth, Luci Feinberg, Aliya Tang, Elvina Wibisono, Ena Selman-Housein, Leyi Cui, Yana Botvinnik, Seojin (Heidi) Yoon, Elizabeth Fields, Chloe Ho, Min Jie Kim, Zixuan Wu FALL 2023
 - (Masters) William Gerard, Xueyi Bu, (Undergraduate) Emre Adabag, Nolan Tremelling SUMMER 2023
 - (Masters) Donghan Kim, Miloni Atal, Srishti Srivastava, William Gerard, Xueyi Bu, Han Lin SPRING 2023
 - (Undergraduate) Rachel Peng, Kofi Meighan, Alekhya Maram, Luci Feinberg, Elvina Wibisono, Eris Gao, Kevin Luo, Habeeba Mansour, Nolan Tremelling, Catherine Ji, Chau “Chloe” Nguyen, Kimiya Shahamat, Danelle Tuchman, Emre Adabag, Yana Botvinnik, Seojin “Heidi” Yoon, Antonio Aranda, Avigayil Helman, Nick Luo, Annie Song, Jessica Zhang, Jennifer Luo, Bonnie Yang, Alekhya Maram, William Xie, Alexander Du, Malik Endsley, Luci Feinberg, Aliya Tang, Elvina Wibisono, Elizabeth Fields, Min Jie Kim, Zixuan Wu, Samarth Agrawal
 - (Masters) William Gerard, Miloni Atal, (Undergraduate) Emre Adabag, Antonio Aranda, Catherine Ji, Kimiya Shahamat FALL 2022
- Harvard College Women in STEM Mentorship Program FALL 2021 – SPRING 2022
- Try AI: “A program designed to introduce early undergraduate students, particularly those identifying as women, Black, Latinx, and/or Indigenous, to research in Artificial Intelligence” FALL 2020

INVITED TALKS, SEMINARS, AND PANELS

- “An Introduction to Robotics Acceleration with Computing Hardware and Systems” JUNE 2026
RoboARCH: Robotics Acceleration with Computing Hardware and Systems at the 2026 IEEE International Conference on Robotics and Automation (ICRA), Vienna, Austria, USA
- “GPU-Accelerated Optimization for Model Predictive Control at the Edge” JUNE 2026
SIAM Conference on Optimization, Edinburgh, Scotland, United Kingdom
- “Optimizing Robotic Systems at all Scales” DEC 2025
École Polytechnique Fédérale de Lausanne Automatic Control Laboratory Seminar, Lausanne, Switzerland
- “Optimizing Robotic Systems at all Scales” NOV 2025
Harvard John A. Paulson School of Engineering Robotics Seminar, Boston, MA, USA
- “Advanced TinyML Initiatives: From TinyML to EdgeAI Systems” SEPT 2025
“Embedded Robotics: Edge Computational Challenges and Opportunities”
SciTinyML 2025: Scientific Use of Machine Learning on Low-Power Devices, Bogota, Colombia
- “Introduction to Trajectory Optimization and Numerical Optimal Control - Parts I and II” JULY 2025
Optimization for Robotics Summer School, Patras, Greece
- “Optimizing Robotic Systems at all Scales” JUNE 2025
Workshop on Fast Motion Planning and Control in the Era of Parallelism at Robotics Science and Systems (RSS), Los Angeles, CA, USA
- “An Introduction to Robotics Acceleration with Computing Hardware and Systems” MAY 2025
“Developing a Roadmap and Whitepaper for the Future of Systems and Architecture for Robotics”
RoboARCH: Robotics Acceleration with Computing Hardware and Systems at the 2025 IEEE International Conference on Robotics and Automation (ICRA), Atlanta, Georgia, USA
- “Optimizing Robotic Systems at all Scales” MAY 2025
Autonomy Talks, Remote
- “Structure Exploitation and Preconditioning for Real-Time GPU-Accelerated Optimization at the Edge” MAR 2025
SIAM Conference on Computational Science and Engineering, Fort Worth, TX, USA
- “Optimizing Robotic Systems at All Scales” FEB 2025
Dartmouth College Computer Science Seminar Series, Hanover, NH, USA
- “Optimizing Robotic Systems at All Scales” FEB 2025
Tufts University Electrical and Computer Engineering Seminar Series, Medford, MA, USA
- “Challenges and Opportunities for Computational Robotics Across Scales” NOV 2024
Intel Neuromorphic Computing Labs Seminar Series, Remote
- “Acceleration of (Nonlinear) Optimization for Robotics through Co-Design: from MCUs to GPUs” OCT 2024
INFORMS Annual Meeting, Seattle, WA, USA
- “Optimization for Robotics through Co-Design” OCT 2024
New York University, Robotics Seminar Series, New York, NY, USA
- “Towards a Comprehensive Package for GPU-Accelerated Trajectory Optimization and Rigid Body Dynamics” SEPT 2024
Northeast Robotics Colloquium (NERC), Amherst, Massachusetts, USA
- “Optimization for Robotics through Co-Design” SEPT 2024
Barnard Computation Science Center: Computing Across Disciplines Talk Series, New York, NY, USA
- “From Convolutional Filters to Computer Vision at the Extreme Edge” JULY 2024
“Tiny Robots: Edge Computational Challenges and Opportunities”
“Sustainable & Responsible TinyML”
Workshop on TinyML for Sustainable Development, Sao Paulo, Brazil
- “Optimizing at All Scales: Edge (Non)linear Model Predictive Control from MCUs to GPUs” JULY 2024
Lightning Talk at the Frontiers of Optimization for Robotics Workshop at Robotics Science and Systems (RSS), Delft, Netherlands
- “Panelist: Sustainable Robotics and Climate Change” MAY 2024
Workshop on Robots and Roboticians in the Age of Climate Change at the IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan
- “Responsible TinyML” MAY 2024
SciTinyML-24: Workshop on Machine Learning on Low-Power Devices: Applications and Advanced Topics, Remote
- “TinyML4D: Scaling Embedded Machine Learning Education in the Developing World - A Hands-On Workshop” MAY 2024
AAAI Spring Symposium on Increasing Diversity in AI Education and Research, San Francisco, CA, USA
- “The Big Impact of TinyML: Embedded Machine Learning at the Extreme Edge” JULY 2023
1st Morocco AI Summer School, Ifrane, Morocco and Remote
- “ML Sensors and the Environmental Impact of TinyML” JULY 2023
“Launching TinyML edX and Long Term Support”
EdgeMLUP-23: Workshop on Widening Access to TinyML Network by Establishing Best Practices in Education, Trieste, Italy

- “Is TinyML Sustainable? Assessing the Environmental Impacts of Machine Learning on Microcontrollers” APR 2023
SciTinyML-23: Workshop on Scientific Use of Machine Learning on Low-Power Devices: Applications and Advanced Topics, Remote
- “GPU Acceleration for Real-Time, Whole-Body, Nonlinear Model Predictive Control” APR 2023
Columbia University Computer Science Faculty Seminar, New York, NY, USA
- “GPU Acceleration for Real-Time, Whole-Body, Nonlinear Model Predictive Control” NOV 2022
University of Waterloo MME Departmental Seminar Series, Remote
- “Machine Learning for HPC” OCT 2022
Energy Efficient High Performance Computing Working Group, Remote
- “Keyword Spotting with Convolutional Neural Networks” JULY 2022
“Introduction to Artificial Intelligence and (Tiny)ML”
EASI-22: The 2022 Edge AI Summer Institute, Remote
- “Convolutions and Transfer Learning for Computer Vision” JULY 2022
“The Future of Machine Learning is Tiny and Bright”
SciTinyML-22 Latam: Latin America Regional Workshop on Scientific Use of Machine Learning on Low-Power Devices Workshop, Remote
- “Data Pre-Processing for Hands-on Keyword Spotting” APR 2022
“Convolutions for Hands-on Computer Vision”
SciTinyML-22 Africa: African Regional Workshop on Scientific Use of Machine Learning on Low-Power Devices, Remote
- “TinyMLedu: Widening Access to TinyML Education and Resources” JAN 2022
TinyML Toronto Meetup, Remote
- “Hardware Acceleration for Realtime Robotics” DEC 2021
Barnard College, New York, NY, USA
- “Introduction to Robot Motion Planning” DEC 2021
Simmons College, Boston, MA, USA
- “TinyMLedu Outreach: Embedded Machine Learning for the Navajo Nation” NOV 2021
TinyML for Good, Remote
- “Hands-on Embedded ML from Theory to Practice: Vision and Audio” OCT 2021
SciTinyML-21: Scientific Use of Machine Learning on Low-Power Devices Workshop, Remote
- “Robust Realtime Model Predictive Control through Co-Design” DEC 2020
Carnegie Mellon University, Robotic Exploration Lab, Remote
- “Custom Accelerator Chips (ASICs) for Robotics / Autonomous Systems” APR 2020
Pillar VC, Boston, MA, USA
- “Accelerating Real Time Model Predictive Control” OCT 2019
Optimus Ride, Boston, MA, USA
- “GPU Acceleration for Robotics” NOV 2018
DUT/MIT Formula Student Driverless, Cambridge, MA, USA

UNIVERSITY COMMITTEES

Dartmouth College

- Scaling AI Systems Faculty Search Committee FALL 2026 – PRESENT
- Improving Healthcare Outcomes through Sensory Technology Cluster Search Committee FALL 2025 – PRESENT
- Computer Science PhD Admission Committee FALL 2025 – PRESENT

Barnard College, Columbia University:

- Barnard College Summer Research Institute Department Representative for Computer Science 2022 – 2025
- Barnard Center for Research on Women (BCRW) Faculty Advisory Board 2022 – 2025
- Barnard College Computer Science Faculty Search Committee FALL 2024 – SPRING 2025
- Columbia University Robotics Faculty Search Committee – Affiliate Member SPRING 2024
- Barnard College Computer Science Tenured Faculty Search Committee FALL 2023

Harvard University:

- Harvard Women in STEM ENVISION Proposal-Writing Competition Judges Panel 2021-2022
- SEAS Committee on Diversity, Inclusion, and Belonging: Post-Baccalaureate Outreach Working Group 2021
- Harvard i3: The Harvard College Innovation Challenge Judges Panel 2021

OUTREACH AND PROFESSIONAL SERVICE

ORGANIZATIONAL LEADERSHIP

- IEEE RAS Technical Committee on Optimization for Robotics [Link]: Co-Chair: 2026 – PRESENT
 - Won the “Blue Ribbon TC Award” in 2023 (formerly “Most Active TC Award”) Associate Co-Chair: 2022 – 2025
 - Previously known as the Technical Committee on Model-Based Optimization for Robotics

- AIEng4D: AI Engineering for Global Impact: *Founding Co-Chair*: 2021 – PRESENT
 - Previously known as the Tiny Machine Learning Open Education Initiative (TinyMLedu) [\[Link\]](#)
- EdgeAI Foundation Academia-Industry Partnership [\[Link\]](#): *Founding (Co-)Chair*: 2024 – PRESENT

WORKSHOP, SUMMER SCHOOL, AND SPECIAL SESSION CO-ORGANIZER

- RoboARCH 2022-26: Workshop on Robotics Acceleration with Computing Hardware and Systems. This workshop was held both as an IEEE/ACM International Symposium on Microarchitecture (MICRO) Workshop [\[Link-22, 23, 24\]](#) and as an IEEE International Conference on Robotics and Automation (ICRA) Workshop [\[Link-25, 26\]](#).
- Workshop on Open Robotics Software. An IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2026 Workshop [\[Link\]](#).
- SciTinyML 2021-25: Scientific Use of Machine Learning on Low-Power Devices. a 5-day hands-on, virtual workshop for university students and professors exploring real-world applications of TinyML and their impact on the developing world. 2021 was run globally with 216 participants from 48 countries, 2022 was run regionally for Africa (187 from 29), Asia (100 from 8), and Latin America (200 from 17), 2023-24 were run globally (418 from 76 and 201 from 44). 2025 shifted to an in-person regional workshop for Latin America in Columbia [\[Link-21, 22, 23, 24, 25\]](#).
- Optimization for Robotics Summer School - 2025. An IEEE RAS-TEP Summer School held at the University of Patras, Greece with 90 participants and over 30 fee waivers and travel awards extended to participants [\[Link\]](#).
- Special Session on Tiny Machine Learning at the 2025 International Joint Conference on Neural Networks [\[Link\]](#).
- Workshop on TinyML for Sustainable Development 2024. An in-person, 5-day, hands-on workshop co-organized and hosted by IBM in Sao Paulo, Brazil focused how TinyML can be leveraged for the benefit of all globally [\[Link\]](#).
- Workshop on Leveraging Models for Contact-Rich Manipulation. An IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2023 Workshop [\[Link\]](#).
- EdgeMLUP 2023: Workshop on Widening Access to TinyML Network by Establishing Best Practices in Education. An in-person, 5-day, hands-on workshop at ICTP for university level educators, focused on the global south, to develop and expand access to embedded machine learning curricula and research programs [\[Link\]](#).
- 3rd On-Device Intelligence Workshop. A Conference on Machine Learning and Systems (MLSys) 2023 Workshop. [\[Link\]](#)
- Building Approachable, Hands-On Embedded Machine Learning Curriculum Using Edge Impulse and Arduino. An AAAI 2023 Conference on Artificial Intelligence Tutorial and Lab Forum [\[Link\]](#).
- Mind the Gap: Opportunities and Challenges in the Transition Between Research and Industry. A Robotics Science and Systems (RSS) 2022 Workshop [\[Link\]](#).
- EASI 2022: the Edge AI Summer Institute. A 3-day, hands-on workshop for high school teachers and students serving the Navajo Nation exploring artificial intelligence through hands-on examples of TinyML. This program was a collaboration between Harvard University, Navajo Technical University and Barnard College, Columbia University [\[Link\]](#).
- CRESTLEX 2021: CReating Effective STem Learning Experiences. A 4-day, hands-on workshop for high school teachers and students serving the Navajo Nation exploring artificial intelligence through hands-on examples of TinyML. This program was a collaboration led by Harvard and Navajo Technical University with support from Google and Edge Impulse [\[Link\]](#).

ASSOCIATE EDITOR / PROGRAM COMMITTEE:

- IEEE Robotics and Automation Letters (RAL) – Humanoids and Animaloids: 2024 – PRESENT
- IEEE International Conference on Robotics and Automation (ICRA): 2026, 27
- IEEE International Conference on Humanoid Robots (Humanoids): 2024-26
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS): 2026 Workshops and Tutorials
- IEEE International Conference on Simulation for Modeling and Programming for Autonomous Robots (SIMPAP): 2025
- TinyML Research Symposium: 2025-26

GRANT PANELIST AND REVIEWER:

- NSF CISE Review Panels: 2024
- IEEE Transactions on Robotics (T-RO): 2023, 2025-26
- IEEE Transactions on Control Systems Technology (TCST): 2023
- IEEE Micro: 2023
- IEEE Robotics and Automation Letters (RAL): 2019
- IEEE International Conference on Robotics and Automation (ICRA): 2021-23, 25-26
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS): 2020, 22-26
- Robotics Science and Systems (RSS): 2023-26
- IEEE Symposium on Safety, Security, and Rescue Robotics (SSRR): 2025
- IEEE Conference on Decision and Control (CDC): 2024
- IFAC Conference on Nonlinear Model Predictive Control (NMPC): 2024
- Conference on Neural Information Processing Systems (NeurIPS): Datasets and Benchmarks Track 2023-25
- ACM Technical Symposium on Computer Science Education (SIGCSE TS): Posters 2023, APC Posters 2024, Papers 2025
- ACM Innovation and Technology in Computer Science Education (ITiCSE): 2022-24

- Learning and Intelligent Optimization Conference (LION): 2025
- IEEE International Conference on Advanced Motion Control (AMC): 2022
- IEEE Integrated STEM Education Conference (ISEC): 2018
- UN IATT STI Forum: 2023

OTHER NOTABLE SERVICE:

- 2026 IEEE-RAS Robot Control TC Best Paper Award Committee
- 2026 IROS Learning Coordinated Behavior in Multi-Robot Teams Session Chair
- 2024 ICRA Reinforcement Learning I Session Chair

PROFESSIONAL SOCIETY MEMBERSHIP

- Institute of Electrical and Electronics Engineers (IEEE)
- IEEE Robotics and Automation Society (IEEE-RAS)
- IEEE-RAS Technical Community (TC) on Model-Based Optimization for Robotics (TCOptRob), TC on Humanoids, TC on Performance Evaluation & Benchmarking, TC on Robot Learning, TC on Robot Ethics
- IEEE Computer Society (IEEE-CS)
- IEEE-CS Technical Community (TC) on Parallel Processing (TCPP)
- IEEE Women in Engineering (IEEE-WiE)
- Association for Computing Machinery (ACM)
- ACM Special Interest Group on Computer Science Education (ACM-SIGCSE)
- Association for the Advancement of Artificial Intelligence (AAAI)
- Institute for Operations Research and the Management Sciences (INFORMS)
- Society for Industrial and Applied Mathematics (SIAM)
- Women in AI & Robotics

ADDITIONAL PROFESSIONAL EXPERIENCE

PILLAR VC: Boston, MA – *Venture Fellow* JAN 2020 - MAY 2020

- Assisted in sourcing and evaluating potential investments with a focus in robotics and AI

MCKINSEY & COMPANY: Boston, MA – *Business Analyst* AUG 2013 - JULY 2015

- Offered promotion to senior associate (designation reserved for top ~5% of business analyst class)
- On multiple projects, played “junior engagement manager” role, project managing newer business analysts
- Led product development strategy for a software product, interfacing between multiple engineering teams and management at the business unit and corporate levels for a prime defense contractor
- Analyzed the value chain, market dynamics, and entry strategy for a new product for a high-tech materials manufacturer
- Led business case and market entry strategy development for a novel aircraft for a major aerospace manufacturer
- Designed a network deployment and capital planning strategy for a national wireless carrier
- Facilitated a culture transformation for a Fortune 500 company including a simultaneous worldwide conference
- Designed a comprehensive economic development and revitalization strategy for a rural area of the United States
- Performed the strategic due diligence that led to a successful large (>\$10Bn) deal between Fortune 500 companies

UNITED STATES DEPARTMENT OF DEFENSE: Fort Meade, MD – *Cryptologic Access Program* MAY 2012 - AUG 2012

- Led an internal consulting project on the use of Cloud-based solutions by developing a Map-Reduce Analytic backed GUI prototype and researching and presenting findings on possible use cases to division management
- Helped lead weekly meetings to coordinate a 15 person development team and ensure code reuse and efficient production
- Designed and built generic and reusable widgets in EXT-JS and the Ozone Widget Framework in order to aid leadership in analyzing current operational metrics and improve operational efficiency and oversight

US GREEN DATA: Cambridge, MA – *Senior Software Engineer* SEPT 2011 - MAY 2012

- Managed the tech team consisting of 4 Harvard Students and 4 professional web developers to ensure collaboration and production as lead web designer and developer in a Django / jQuery environment
- Evaluated the web development direction, product line and client requirements with the CEO and other company leaders
- Provided system administrator function to maintain and improve the web architecture, security and design